SANKET SAHA

Software Engineer · Databricks · Spark · Kafka · AWS · Spring · React
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SUMMARY

11 years of experience in software engineering, having contributed across multiple domains — from full-stack development to backend systems, and for the past five years, as a data engineer. I am currently employed as a Data Engineer at JPMorgan Chase. Throughout my career, identifying and solving complex problems has been the driving force behind my growth in this field.
Going forward, I aspire to solve problems at scale and to build and own a software product of my own, continually seeking challenges that push me beyond my comfort zone. Outside of work, I enjoy a slow-paced life — hiking, good food, and spending time with family.

TECHNICAL SKILLS

- **Data Engineering:** Databricks, Apache Spark, PySpark, Kafka, Kinesis, Spring Batch, Delta Lake, Medallion Architecture, Streaming, ETL/ELT, Parquet, Unity Catalog.
- **Cloud & Infra:** AWS (ECS, EKS, S3, Lambda, Kinesis, CloudWatch, Glacier, IAM), Docker, Kubernetes, Terraform, Helm, DABS, Jenkins, Github, GitHub Actions
- **AI Agents:** Google ADK, LangChain, MCP (Model Context Protocol), OpenClaw, Pydantic , Mem0 , Claude , Open AI
- **Languages:** Python, Java 11, SQL, TypeScript, Scala
- **Backend & DB:** Spring Boot, Microservices, MySQL, Oracle, DynamoDB, MongoDB , FastAPI
- **Observability:** Spark UI, DAG Analysis, Splunk, Grafana, Prometheus, Spark Streaming Listener

EXPERIENCE

Senior Data Engineer (Software Engineer III) | JP Morgan Chase
Sep 2024 – Present · Hyderabad, India

- **Built observability platform across 100+ live Databricks streaming jobs** — Spark Streaming Listener metrics surfaced into a live ops dashboard. Leaders and ops teams now have real-time visibility into lag, throughput, errors, and job health. Eliminated all manual monitoring.
- **Extended the observability platform** with **LLM agents** (Google ADK + Mem0) that analyse Spark event metrics, diagnose anomalies, and recommend fixes — with incident memory across runs, turning raw monitoring into actionable triage.
- **Designed and implemented - E**nd-to-End Databricks projects from scratch using DABS, creating optimized workflows and Delta Live Tables (DLT) pipelines. Leveraged CI/CD for seamless deployment, ensuring scalable and efficient data processing.
- **Reduced pipeline runtime by 50% (50 min → 23 min)** via systematic Spark UI and DAG analysis — right-sizing partitions, tasks, and clusters. Outcomes formalised into a team-wide performance guide.
- **Cut ingestion boilerplate by 90% (1,000+ lines → <100)** — designed a reusable Maven JAR encapsulating core ingestion logic. New source onboarding is now fast and consistent across all applications.
- **Redesigned Kafka ingestion to load directly into Databricks raw tables** — removing an intermediate ECS layer, lowering infrastructure cost and pipeline complexity.
- **Delivered multi-source Lakehouse platform** — orchestrating Kafka streams and REST API sources across JPMC systems into S3, then Delta tables (Bronze/Silver/Gold), with cost controls via S3 Glacier tiering and DABS-based CI/CD deployment.

Software Engineering Lead | Optum / UnitedHealth Group
Jan 2021 – Sep 2024 · Hyderabad, India

- **Processed 100,000 transactions in under 3 minutes** via Kafka + Spring Batch pipeline — 4x faster than the prior batch system.
- **Reusable Kafka and Object Storage libraries adopted by 30+ microservices** — standardised data processing and cut integration effort across the organisation.
- Resolved critical ETL and streaming bottlenecks; built AOP-based structured logging that cut incident resolution time significantly.
- Mentored 4 engineers, led code reviews, and drove clean architecture adoption across the data engineering team.

Consultant — Data Engineer / Full Stack | Deloitte USI
Apr 2018 – Dec 2020 · Hyderabad, India

- **Reduced SAP Order Sync failures by 90%** — resolved race conditions and data inconsistencies eliminating recurring SLA breaches.
- **Cut manual translation effort by 70%** with a Python automation POC — established a reusable pattern for multilingual processing.
- Delivered complex features (**Full Palletization, Order Management, Market Validation**) independently for energy sector clients.

System Engineer | Tata Consultancy Services (TCS)
Sep 2015 – Apr 2018 · Hyderabad, India

- Redesigned **cross-channel CCV system** with server-side validation, improving security and data consistency for a major banking client.
- Led system impact analysis and enforced coding **standards, reducing rework** across a distributed development team.

ADDITIONAL CONTRIBUTION

- Built a full-stack School Management System (Java + React + MySQL) — 150+ students, live attendance, fee tracking, WhatsApp parent notifications, Metabase analytics, Weekly planner, AI-Driven Smart Review.
- Built AI SDK to integrate naturally with different systems (School & Personal) .
- Self-hosted home server running secure storage, media streaming, and home automation — hands-on DevOps beyond enterprise context.
- Hackathon participant (UHG, Deloitte). 100+ LeetCode. Hacktoberfest open-source contributor.

EDUCATION

B.Tech in Information Technology · GITAM University · *2011 – 2015*

CERTIFICATIONS

- **Databricks Certified Data Engineer Associate** — Databricks · *Sept 2026*
- **AWS Certified Cloud Practitioner (CLF-C02)** — AWS · *July 2026*
- **Oracle Certified Professional, Java SE 6** — Oracle · *Nov 2014*